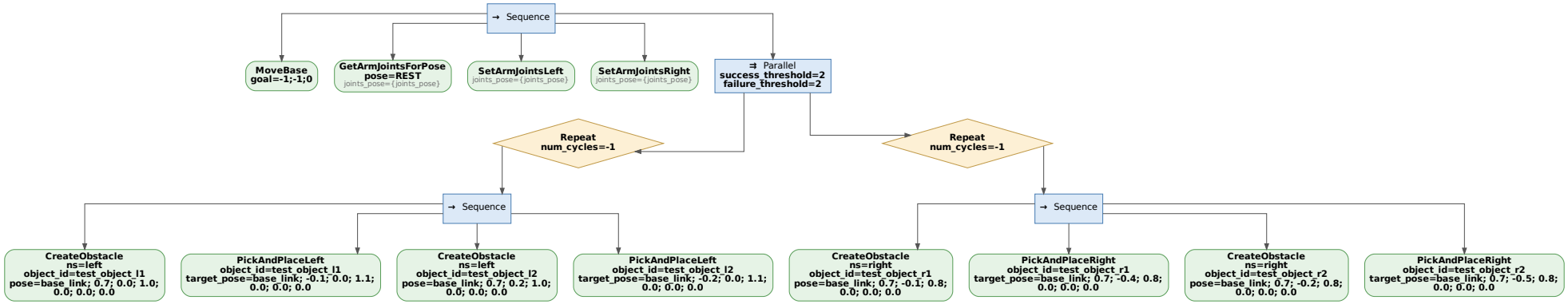
Behavior tree



Mission requirement (mrq)

The mission describes a bimanual pick-and-place task that moves to a start pose, parks both arms in a rest configuration, and then runs continuous left and right object handling in parallel.
First, the robot will execute the following actions (MoveBase, GetArmJointsForPose, SetArmJointsLeft, SetArmJointsRight, Parallel) in order and stop the sequence if any action fails. It will run the custom action MoveBase with the goal set to coordinates -1, -1, 0. It will run the custom action GetArmJointsForPose with the pose set to REST and will store the resulting joints_pose. It will run the custom action SetArmJointsLeft using the stored joints_pose. It will run the custom action SetArmJointsRight using the stored joints_pose. It will execute the following actions [Repeat, Repeat] simultaneously. If either a success threshold is reached or all actions succeed, the execution was successful. Else if any action fails, the execution fails.
Next, it will execute Repeat for a set number of times or infinitely. It will execute the following actions (CreateObstacle, PickAndPlaceLeft, CreateObstacle, PickAndPlaceLeft) in order and stop the sequence if any action fails. It will execute CreateObstacle in the left namespace to create object test_object_l1 at base_link with position 0.7, 0.0, 1.0 and orientation 0.0, 0.0, 0.0. It will execute PickAndPlaceLeft for object test_object_l1 to the target pose at base_link with position -0.1, 0.0, 1.1 and orientation 0.0, 0.0, 0.0. It will execute CreateObstacle in the left namespace to create object test_object_l2 at base_link with position 0.7, 0.2, 1.0 and orientation 0.0, 0.0, 0.0. It will execute PickAndPlaceLeft for object test_object_l2 to the target pose at base_link with position -0.2, 0.0, 1.1 and orientation 0.0, 0.0, 0.0. It will execute this sequence for a set number of times or infinitely.
Finally, it will execute Repeat for a set number of times or infinitely. It will execute the following actions (CreateObstacle, PickAndPlaceRight, CreateObstacle, PickAndPlaceRight) in order and stop the sequence if any action fails. It will execute CreateObstacle in the right namespace to create object test_object_r1 at base_link with position 0.7, -0.1, 0.8 and orientation 0.0, 0.0, 0.0. It will execute PickAndPlaceRight for object test_object_r1 to the target pose at base_link with position 0.7, -0.4, 0.8 and orientation 0.0, 0.0, 0.0. It will execute CreateObstacle in the right namespace to create object test_object_r2 at base_link with position 0.7, -0.2, 0.8 and orientation 0.0, 0.0, 0.0. It will execute PickAndPlaceRight for object test_object_r2 to the target pose at base_link with position 0.7, -0.5, 0.8 and orientation 0.0, 0.0, 0.0. It will execute this sequence for a set number of times or infinitely.